



**Verified Carbon
Standard**

INNER MONGOLIA WUDAOGOU 50.25MW WIND POWER PROJECT

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Project Title	Inner Mongolia Wudaogou 50.25MW Wind Power Project
Version	02
Report ID	-
Date of Issue	03-July-2022
Project ID	150
Monitoring Period	01-May-2012 to 31-Dec-2016 (Both days included)
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CONTENTS

1	PROJECT DETAILS.....	3
1.1	Summary Description of the Implementation Status of the Project	3
1.2	Sectoral Scope and Project Type	3
1.3	Project Proponent	3
1.4	Other Entities Involved in the Project	4
1.5	Project Start Date	4
1.6	Project Crediting Period	4
1.7	Project Location	4
1.8	Title and Reference of Methodology	5
1.9	Participation under other GHG Programs.....	5
1.10	Other Forms of Credit.....	6
1.11	Sustainable Development Contributions	6
2	SAFEGUARDS	9
2.1	No Net Harm	9
2.2	Local Stakeholder Consultation	9
2.3	AFOLU-Specific Safeguards	11
3	IMPLEMENTATION STATUS	11
3.1	Implementation Status of the Project Activity	11
3.2	Deviations	11
3.3	Grouped Projects	12
4	DATA AND PARAMETERS.....	12
4.1	Data and Parameters Available at Validation	12
4.2	Data and Parameters Monitored.....	13
4.3	Monitoring Plan	14
5	QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	17
5.1	Baseline Emissions	18
5.2	Project Emissions	23
5.3	Leakage.....	23
5.4	Net GHG Emission Reductions and Removals.....	23

1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Inner Mongolia Wudaogou 50.25MW Wind Power Project is a grid connected renewable energy project located in Inner Mongolia Autonomous Region, Northeast China. The objective of the project is to generate electricity using wind power generation technology and to sell into the Power Grid. The project involves the installation of 67 sets of 750 kW wind turbines, which amounts to a total installed capacity of 50.25MW.

The project is expected to supply 124.66GWh renewable electricity annually to the power grid during the 20 years lifetime. The project achieves CO₂ emission reduction by replacing electricity generated by fossil fuel fired power plant connected into Northeast China Power Grid (NEPG). The estimated annual emission reduction of the project is 142,848 tCO_{2e}.

The project was implemented in consistency with the registered PD. The project began to construct on 08-April-2006 and started fully operation since 01-Jan-2007¹.

This is the 2nd monitoring period from 01-May-2012 to 31-Dec-2016 under VCS program for the project. The total GHG emission reductions generated in this monitoring period are 551,292 tCO_{2e}.

1.2 Sectoral Scope and Project Type

Sectoral scope: 1 energy industries (Renewable sources).

Project type: Grid-connected wind power project.

The project is not a grouped project.

1.3 Project Proponent

Organization name	Chifeng Xinsheng Wind Power Co., Ltd.
Contact person	Haihan Wang
Title	Vice Manager of Marketing Department
Address	Yihong town, Wengniute Banner, Chifeng City, Inner Mongolia

¹ All the 67 sets wind turbine generators have been connected to the power grid on 31-Dec-2006 and started power generation on 01-Jan-2007, which is also the start date of VCS crediting period of the project. Therefore, the fully operation start date of the project is confirmed to be 01-Jan-2007.

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1.4 Other Entities Involved in the Project

Organization name	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.
Role in the Project	Consultant
Contact person	Shanshan Yu
Title	General Manager
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1.5 Project Start Date

01-Jan-2007 (Fully operation start date, please refer to footnote 1 above for details).

1.6 Project Crediting Period

The 1st crediting period of the VCS project was 10 years from 01-Jan-2007 to 31-Dec-2016.

1.7 Project Location

The project is located in Wengniute County, East Inner Mongolia, China. Its geographical coordinates are north latitude 42°39'-42°42' and east longitude 117°54'-118°01'. The figure below shows the location of the project site.

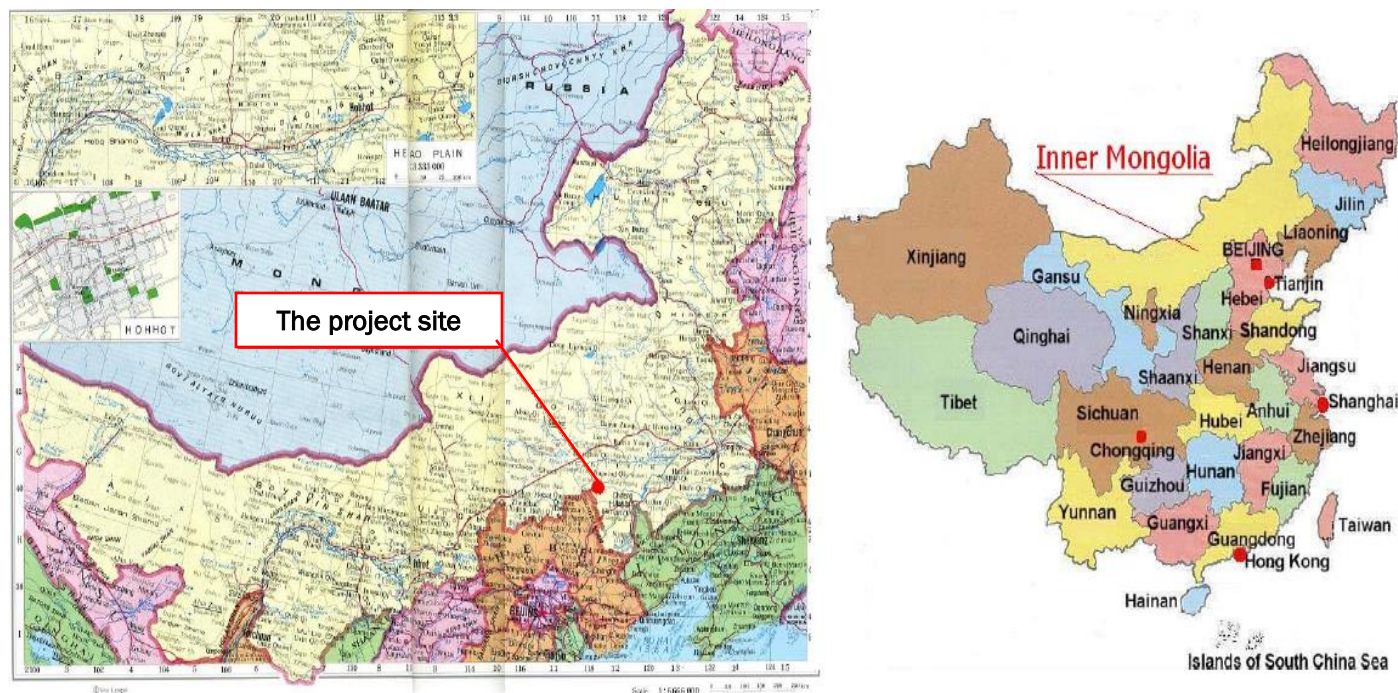


Figure 1 Location of the Project

1.8 Title and Reference of Methodology

Applied methodology: ACM0002 (Version 6): Consolidated baseline methodology for grid-connected electricity generation from renewable sources.

Applied tools: Tool for the Demonstration and Assessment of Additionality (version 03).

The applied methodology and tool can be found from the following link:

<https://cdm.unfccc.int/methodologies/PAMethodologies/approved>

1.9 Participation under other GHG Programs

The project is a registered CDM project with reference No. 1487, the detailed information can be found at <https://cdm.unfccc.int/Projects/DB/DNV-CUK1198674466.99/view>.

The emission reductions achieved during period 05-Jul-2008 to 30-Apr-2012 have been issued as CERs under CDM program. The first monitoring period under VCS program was from 01-Jan-2007 to 30-Jun-2008, with VCUs successfully issued. This is the 2nd monitoring period under VCS program from 01-May-2012 to 31-Dec-2016 for the project. All emission reductions during this monitoring period have not and will not seek CDM CER issuance. Emission reductions during this monitoring period will only seek issuance under VCS program.

1.10 Other Forms of Credit

Emission Trading Programs and Other Binding Limits

China has a national emissions trading scheme only cover the high-emission industries, such as thermal power generation, petrochemical, chemical, building materials, iron and steel, non-ferrous, paper, aviation and other key emission industries that emitted at least 26,000 tons of CO₂e/year, not including renewable project². Thus, the project proponent: Chifeng Xinsheng Wind Power Co., Ltd., as an enterprise for renewable energy investment, is not included in the compliance entity list by China national Emission Trading Scheme (ETS). Moreover, the project has not been registered as a CCER (Chinese Certified Emission Reductions) project in China, thus it is not eligible for emission reductions transaction under the China's ETS.

Therefore, the project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading. The net GHG emission reductions generated during this monitoring period have not been used for compliance under such programs or mechanisms.

Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit, including renewable energy certificates, during this monitoring period.

1.11 Sustainable Development Contributions

The project activity is a renewable wind power project located in Inner Mongolia Autonomous Region, Northeast China.



The project utilizes wind resources to generate and supply 481,102.502MWh renewable electricity to the power grid during this monitoring period, which contributes to SDG 7.



The project provides 22 long-term decent job opportunities for local residents, which has a positive effect on the local economy and contributes to SDG 8.

² http://www.mee.gov.cn/xxgk2018/xxgk/xxgk05/202103/t20210330_826728.html



The renewable electricity generated by the project replaces electricity generation by fossil-fuel power plants of the power grid, which reduces 551,292 tCO₂e GHG emission during this monitoring period and contributes to SDG 13.

Since the project contributions with all impacts are not included in previously approved VCS monitoring report or reported in Sustainable Development Contribution Reports, all impacts during the previous monitoring periods from 01-Jan-2007 to 30-Apr-2012 has not been calculated in the cumulative impact. Thus, the cumulative impacts in project lifetime are same as the one in this monitoring period.

Please refer to section 4 and section 5 of this monitoring report for the evidence of the project's contribution to SD 7 and SDG 13. And the evidence for SDG 8 is submitted as a separate document to Verra due to confidentiality requirement of the project proponent.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	Renewable energy supplied to the power grid	Implemented activities to increase	481,102.502MWh of electricity from renewable sources has been supplied to the power grid during the reporting period	An accumulated 481,102.502MWh of electricity from renewable sources has been supplied to the power grid by end of the reporting period
2)	8.5	Numbers of decent job opportunities provided by implementation of the project	Implemented activities to increase	22 people including 6 female employees have been employed by the project as long-term employee which have an income rate higher than the local average level.	22 people have been employed by the project as long-term employee by end of the reporting period
3)	13.0	Tonnes of greenhouse gas emissions avoided	Implemented activities to increase	The project has achieved a total GHG emission reduction of 551,292 tCO ₂ e during the reporting period.	An accumulated 551,292 tCO ₂ e of GHG emission reduction have been achieved by the project by end of the reporting period

2 SAFEGUARDS

2.1 No Net Harm

The Environmental Impact Assessment (EIA) of the project has been conducted and approved by local government officials. The EIA report has assessed every possible aspect of environmental impacts of the project and recommended corresponding measures, where applicable. The environmental impacts and corresponding mitigation measures during operation have been discussed in the registered PD. No negative environmental impacts have been identified.

Furthermore, the project makes positive contribution to the sustainable development as described in section 1.11 of this report.

In conclusion, no net harm on local environment and social community has been detected for the project.

2.2 Local Stakeholder Consultation

LSC prior to the project implementation

Local stakeholder consultation has been conducted in July 2006. All comments received have been well addressed. Please refer to section E of the registered PD (version 3.1 dated 14-Aug-2012) of the project for detailed information.

LSC during the operation period

Communications with Local stakeholders have been carried out at periodic intervals. Key implementation schedules and changes of the project have been communicated to the local authority, the neighbourhood committee, and local residents. The comments and suggestions from residents were collected by the local authority, if any.

The project owner carried out questionnaire survey for the local stakeholders to collect the relevant comments and suggestions every two years, e.g., in June 2012, July 2014 and May 2016 during this monitoring period. The local authority also conducts spot checks on the implementation of the project at periodic intervals as per relevant regulations.

The following people are considered as the stakeholders of this project during the implementation stage.

- Residence of the nearby village.
- Staff of the project.
- Relevant administrative staff of the local government.

The project owner designed a questionnaire to collect the comments of relevant stakeholders and the major question issues are shown as below:

1. Are you satisfied with the current environmental quality of the project location?
2. What do you think are the main environmental problems in the current location of the project?
3. In your opinion, which environmental protection measures need to be strengthened during the operation period of this project?
4. What impact do you think this project activity may have on local employment and residents' income?
5. Impact on the local economic development.
6. Attitude on implementation of the project.

During each survey, 30 questionnaires were been handed in total and all of them have been received successfully from the stakeholder representatives, respectively from the local government, and surrounding villages. The representatives covered different gender, different ages, different occupations and different education levels. The detail information is shown as below:

Item		Percentage		
		June 2012	July 2014	May 2016
Amount of stakeholders surveyed	Male	42.67%	50%	58.67%
	Female	57.33%	50%	41.33%
Age	<25	5.00%	6.67%	6.67%
	25-45	43.33%	33.33%	43.33%
	45-55	31.67%	43.33%	30.00%
	>55	20.00%	16.67%	20.00%
Education	Junior high school or below	53.33%	55%	30.00%
	Senior high school	43.33%	45%	45.00%
	College or above	3.33%	0%	25.00%
Occupation	Worker	23.33%	10.00%	20.00%
	Farmer	43.33%	53.33%	50.00%
	Management personnel	16.67%	10.00%	6.67%
	Unspecified	16.67%	20.00%	23.33%

All local stakeholders are supportive to the implementation of the project, believing that the Project help improve the life of local people and promote local economic development without adverse environmental impact. There have been no negative comments received for the project so far.

In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.

2.3 AFOLU-Specific Safeguards

The project is a non-AFOLU project, this section is not required.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

67 sets of 750 kW wind turbines (Type S48/750) have been installed by the project, which amounts to a total installed capacity of 50.25MW. Each wind turbine has a 690V-to-35kV transformer, from which a 35kV line links into the on-site 220kV switchgear at the substations established in the project site. By the 220kV line, the electricity generated by the project is delivered to the power grid.

During this monitoring period, the project was implemented in accordance with the registered VCS PD. The Project has operated without any accidental or emergency events that might impact the GHG emission reductions or removals and monitoring.

The emission reductions in the monitoring period from 01-Jan-2007 to 30-Jun-2008 have been issued as VCUs under VCS program and 05-Jul-2008 to 30-Apr-2012 have been issued as CERs under CDM program. During the intervals of 01-Jul-2008 to 04-Jul-2008 (both days included), the project has operated normally without any emergency. The project proponent voluntarily gives up the emission reductions achieved in the period from 01-Jul-2008 to 04-Jul-2008 (Both days included) and will not seek issuance of the emission reductions under any other emission trading programs.

3.2 Deviations

3.2.1 Methodology Deviations

There is no methodology deviation occurred during this monitoring period.

3.2.2 Project Description Deviations

The project is registered under VCS Version 2007.1 and completed validation before 19/03/2020. thus, it remains eligible to apply the crediting period requirements under VCS

version 2007.1 which shall be a maximum of ten years and may be renewed at most twice, so the 1st crediting period of the project should be updated from “01-Jan-2007 to 30-June-2008” to “01-Jan-2007 to 31-Dec-2016”.

The deviation of the crediting period does not impact the applicability of the methodology, additionality or the appropriateness of the baseline scenario.

A revised monitoring plan has been approved by CDM EB on 22-July-2010. According to the revised monitoring plan, the electricity exported to and imported from the grid through the main line are measured by the revenue meter (with accuracy degree of 0.2S) installed at the main transformer of the main line. The electricity imported from the grid through the 10kV backup line is monitored by the meter (with accuracy degree of 1.0) installed on the 10kV backup line.

A PRC has been approved by CDM EB on 27-Dec-2012 due to that the actual accuracy degree of the backup line meter is 2.0 which is lower than that stated in the revised monitoring plan approved by CDM EB on 22-July-2010. According to validation opinion on the PRC, the PRC is applied to the monitoring periods from 01-July-2011 onwards. Thus, the electricity imported from the grid through the 10kV backup line (EG_{backup}) during this monitoring period is calculated as follow:

$$EG_{\text{backup}} = \text{measured value} + \text{measured value} * 0.01.$$

During this monitoring period, the monitoring activity was carried out in compliance with the monitoring plan contained in the approved revised PDD. Details of the data and parameters monitored please refer to Section 4 below.

The deviation of the monitoring plan has no impacts on the applicability of the methodology, additionality, or the appropriateness of the baseline scenario as well as the accuracy and conservativeness of monitoring plan and ER calculation. There is no other project description deviation occurred in this monitoring period or a previous monitoring period.

3.3 Grouped Projects

Not applicable as this is not a grouped project.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	EF _y
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor of the power grid in year y
Source of data	Registered PD

Value applied	1.1459
Justification of choice of data or description of measurement methods and procedures applied	Determined ex-ante and fixed for the 1 st crediting period
Purpose of Data	Calculation of baseline emissions
Comments	-

4.2 Data and Parameters Monitored

Data / Parameter	EG_y		
Data unit	MWh		
Description	Net electricity delivered to the grid by the project		
Source of data	Measurement of electricity meters M1 and M		
Description of measurement methods and procedures to be applied	The metering equipments are properly calibrated annually according to the requirement from Technical administrative code of electric energy metering (DL/T448). The revenue meter M1 (accuracy degree is 0.2S) is installed at the main transformer on the main line, which measures the electricity exported to and imported from the grid through the main line. The meter M installed on the 10kV line (accuracy degree is 2.0) is used to monitor electricity imported from the grid through the 10 kV backup line. The readings of electricity meters are continuously measured and monthly recorded.		
Frequency of monitoring/recording	Continuously measured and monthly recorded		
Value monitored	481,102.502		
Monitoring equipment	Monitoring equipment	Revenue meter M1	Backup line meter M
	Type	ZMD402	DTS8666
	Serial No.	86693424	00605518
	Accuracy	0.2S	2.0
	Calibration frequency	Annually	Annually
	Calibration date	10/02/2012 08/02/2013 06/02/2014	10/02/2012 08/02/2013 06/02/2014

		03/02/2015 01/02/2016	03/02/2015 01/02/2016
	Calibration validity	09/02/2013 07/02/2014 05/02/2015 02/02/2016 31/01/2017	09/02/2013 07/02/2014 05/02/2015 02/02/2016 31/01/2017
QA/QC procedures to be applied	The project operator is responsible for recording this set of data. Electricity sales receipts will also be obtained for double check. Revenue Meter M1 (with accuracy degree of 0.2S) and 10kV backup line meter M (with accuracy degree of 2.0) are calibrated annually by certificated institute. Data will be archived for 2 years following the end of the crediting period by means of electronic and paper backup.		
Purpose of the data	Calculation of baseline emissions		
Calculation method	$EG_y = \text{Electricity exported to the grid through main line } (EG_{ex}) - \text{electricity imported from the grid through the main line } (EG_{im}) - \text{electricity imported from the grid through the 10kV backup line } (EG_{backup})$ i.e. $EG_y = EG_{ex} - EG_{im} - EG_{backup}$		
Comments	-		

4.3 Monitoring Plan

Organizational structure

This monitoring plan is implemented by professional staff authorized by the owner of the project. The management structure is illustrated as follows:

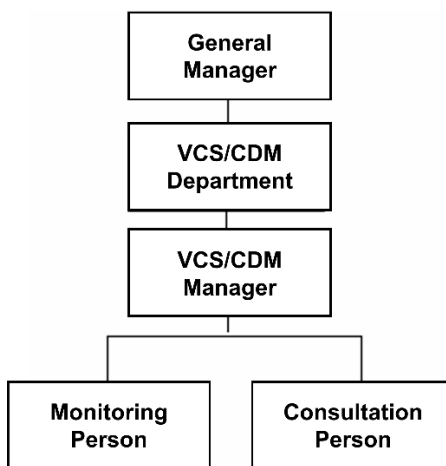


Figure 2 Organizational structure

The Management Group of the project must maintain credible, transparent, and adequate data estimation, measurement, collection, and tracking systems to maintain the information required for an audit of an emission reduction project.

The Management Group consists of professional staff authorized by the owner of the project, i.e. Chifeng Xinsheng Wind Power Co. Ltd.

The Management Group is responsible for the management of the project, which specifically includes selecting a CDM/VCS consultant company, providing support to the selected consultant company, assisting the selected consultant company in selecting VCU buyer and determine VCU price, assisting the selected consultant company in applying for China DNA approval, providing support in the due diligence of the selected buyer, providing support in the CDM/VCS validation and registration by a selected VVB, and providing support in the verification by a selected VVB.

The General Manager is in general control and makes key decisions. The VCS/CDM Department is responsible for specific tasks in monitoring and execution of decisions. Specifically, the VCS/CDM Manager is responsible for the daily operation of the VCS/CDM Department, and contact with VVB and Verra to support their work in validation, registration, verification, and certification. The procedure is illustrated in Section 5 of registered PD Annex.4. He is also responsible for Monitoring Data adjustment and settlement of Data uncertainties, together with local electric power company, and/or VVB. The procedure is illustrated in Section 3 of registered PD Annex.4.

Monitoring equipment

The revenue meter M1 is installed at the main transformer of the main line (with accuracy degree of 0.2S) which measures the electricity exported to and imported from the grid through the main line.

Meter M (accuracy class is 2.0) installed at the 10kV backup line which measures the electricity imported from the grid through the 10kV backup line.

The net electricity supplied to the grid is obtained through the electricity exported to the grid subtract the electricity imported from the grid through the main line and 10kV backup line. The relationship for the parameters shows as follow formula:

$$EG_y = EG_{ex} - EG_{im} - EG_{backup}$$

The installation of electricity meters is shown in the figure below:

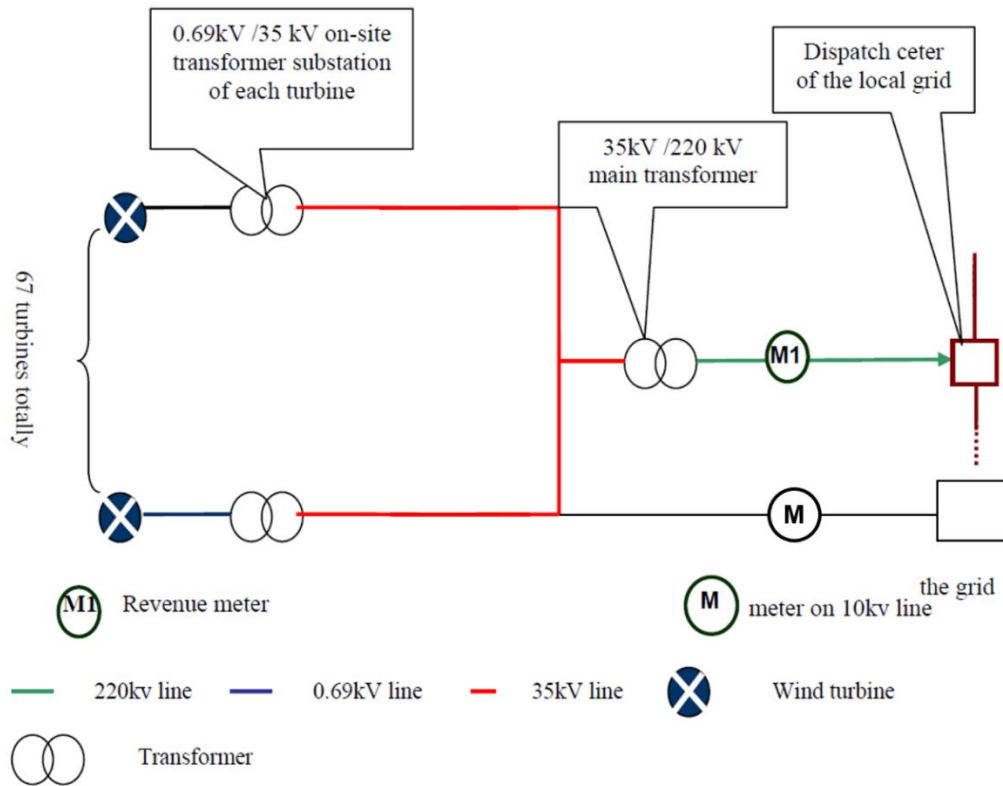


Figure 3 Installation of electricity meters

Data collection

The designated personnel of the project proponent recorded the meter's readings of electricity exported through the main line and electricity imported through 10kV backup line at 24:00 on the last day of every month, while the meter's readings for electricity imported through the main line were recorded at 00:00 on one day from 1st to 10th of each month determined by the grid company. The meter's readings for electricity imported through the main line at 00:00 of 1st January of every year were also recorded. After confirming the readings by both sides, the grid company issues the sales receipts based on the checked electricity amount. The sales receipts are used for double checking purpose to ensure consistency.

Data are archived for 2 years following the end of the crediting period by means of electronic and paper backup.

Monitoring Practice and Handling of Emergency

Should reading of the meters be inaccurate by more than the allowable error, or otherwise functioned improperly, the electricity supplied to the grid by the project shall be determined by:

- 1) First, by reading the self-carried meters of wind turbines, unless a test by either party reveals they are inaccurate;
- 2) If the self-carried meters of wind turbines are not with acceptable limits of accuracy or

are otherwise performing improperly, the project owner and the grid company shall jointly prepare an estimate of the correct reading; and

3) If the project owner and the grid company fail to agree the estimate of the correct reading, then the matter will be referred for arbitration according to agreed procedures.

Data Management System

In order to facilitate verification, all data relevant to monitoring are carefully kept and stored by the project owner.

The following table below outlines the key documents relevant to monitoring and verification of the emission reductions of the project.

No.	Document	Source
F-1	PDD, including the electronic spreadsheets and supporting documentation	project owner
F-2	The monitoring report on the net electricity supplied to the power grid	project owner
F-3	Report on maintenance and calibration of metering equipment	project owner

QA/QC and emergency procedure

An agreement should be signed between the project owner and the grid company that defines the metering arrangements and the required quality control procedures to ensure accuracy.

- The metering equipments are properly calibrated annually according to the requirement from Technical administrative code of electric energy metering (DL/T448).
- After calibration the meters are jointly inspected and sealed on behalf of the parties concerned and shall not be accessible by either party except in the presence of the other party or its accredited representatives.
- The meters installed will be tested by the qualified metrical organization co-authorized by the grid company and the PP within 10 days after:
 - 1). The detection of a difference larger than the allowable error in the reading of the meter, when considering the reactive loss of electrical wire;
 - 2). The repair of all or part of the meter caused by the failure of one or more parts to operate in accordance with the specifications;
 - 3). If any errors are detected, the party owning the meter shall repair, recalibrate or replace the meter and give the other party sufficient notice to allow a representative to attend during any corrective activity.;
- The electricity recorded by the meter will suffice for the purpose of billing and emission reduction verification as long as the error in the meter is within the permissible limits.

During this monitoring period, no emergency occurred.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

As per the registered PD, the baseline emission of the project is calculated as below:

$$BE_y = EG_y \times EF_y$$

Where,

BE_y Baseline emissions in year y (tCO₂e)

EG_y Net electricity delivered to the grid by the project (MWh)

EF_y Baseline emission factor in the year y (tCO₂e/MWh)

The baseline emissions of the project are calculated as follows:

Table 2 Calculation of Baseline emissions

Year	EG _y (MWh)	EF _y (tCO ₂ /MWh)	BE _y (tCO ₂)
2012 01-May-2012 to 31-Dec-2012	65,429.707	1.1459	74,975
2013	118,119.154	1.1459	135,352
2014	99,590.568	1.1459	114,120
2015	94,923.218	1.1459	108,772
2016	103,039.855	1.1459	118,073
Total	481,102.502	-	551,292

Table 3 Monitored values of electricity exported through the main line (EG_{ex}) and electricity imported from 10kV backup line (EG_{backup})

Period		EG _{ex} (MWh)			EG _{backup} (MWh)		
		Monthly Readings from M1	Values from sales receipts	Conservative Quantity	Monthly readings from meter M	Values from sales receipts	Conservative Quantity
From	To	A	B	C=min(A,B)	G	H	I=1.01*max(G,H)

01/05/2012	31/05/2012	9,701.832	9,701.832	9,701.832	0.000	0.000	0.000
01/06/2012	30/06/2012	6,810.000	6,810.000	6,810.000	0.720	0.720	0.727
01/07/2012	31/07/2012	4,850.000	4,850.000	4,850.000	0.000	0.000	0.000
01/08/2012	31/08/2012	3,327.296	3,327.296	3,327.296	0.840	0.840	0.848
01/09/2012	30/09/2012	3,816.935	3,816.935	3,816.935	0.000	0.000	0.000
01/10/2012	31/10/2012	11,715.578	11,715.578	11,715.578	0.000	0.000	0.000
01/11/2012	30/11/2012	13,195.766	13,195.766	13,195.766	0.000	0.000	0.000
01/12/2012	31/12/2012	12,548.489	12,548.489	12,548.489	0.000	0.000	0.000
2012 Subtotal		-	-	65,965.896	-	-	1.576
01/01/2013	31/01/2013	13,378.072	13,378.072	13,378.072	0.000	0.000	0.000
01/02/2013	28/02/2013	14,552.025	14,552.025	14,552.025	0.000	0.000	0.000
01/03/2013	31/03/2013	8,747.404	8,747.404	8,747.404	0.000	0.000	0.000
01/04/2013	30/04/2013	11,692.875	11,692.875	11,692.875	0.000	0.000	0.000
01/05/2013	31/05/2013	8,228.554	8,228.554	8,228.554	1.950	1.950	1.970
01/06/2013	30/06/2013	5,663.662	5,663.662	5,663.662	0.000	0.000	0.000
01/07/2013	31/07/2013	5,393.200	5,393.200	5,393.200	0.000	0.000	0.000
01/08/2013	31/08/2013	7,280.570	7,280.570	7,280.570	0.000	0.000	0.000
01/09/2013	30/09/2013	7,756.154	7,756.154	7,756.154	0.000	0.000	0.000
01/10/2013	31/10/2013	10,315.839	10,315.839	10,315.839	1.250	1.250	1.263
01/11/2013	30/11/2013	14,240.272	14,240.272	14,240.272	0.000	0.000	0.000
01/12/2013	31/12/2013	11,495.251	11,495.251	11,495.251	0.000	0.000	0.000
2013 Subtotal		-	-	118,743.878	-	-	3.232
01/01/2014	31/01/2014	15,286.382	15,286.382	15,286.382	1.330	1.330	1.343
01/02/2014	28/02/2014	5,323.896	5,323.896	5,323.896	0.000	0.000	0.000
01/03/2014	31/03/2014	7,944.352	7,944.352	7,944.352	0.000	0.000	0.000
01/04/2014	30/04/2014	7,449.578	7,449.578	7,449.578	0.000	0.000	0.000
01/05/2014	31/05/2014	11,772.706	11,772.706	11,772.706	0.000	0.000	0.000
01/06/2014	30/06/2014	2,856.188	2,856.188	2,856.188	0.000	0.000	0.000
01/07/2014	31/07/2014	6,218.833	6,218.833	6,218.833	0.780	0.780	0.788
01/08/2014	31/08/2014	2,202.992	2,202.992	2,202.992	0.000	0.000	0.000
01/09/2014	30/09/2014	4,795.591	4,795.591	4,795.591	0.000	0.000	0.000
01/10/2014	31/10/2014	7,958.828	7,958.828	7,958.828	0.000	0.000	0.000
01/11/2014	30/11/2014	12,850.234	12,850.234	12,850.234	0.000	0.000	0.000
01/12/2014	31/12/2014	15,777.479	15,777.479	15,777.479	0.000	0.000	0.000

2014 Subtotal		-	-	100,437.059	-	-	2.131
01/01/2015	31/01/2015	11,655.849	11,655.849	11,655.849	0.000	0.000	0.000
01/02/2015	28/02/2015	7,547.157	7,547.157	7,547.157	0.000	0.000	0.000
01/03/2015	31/03/2015	12,055.975	12,055.975	12,055.975	0.000	0.000	0.000
01/04/2015	30/04/2015	10,481.015	10,481.015	10,481.015	0.000	0.000	0.000
01/05/2015	31/05/2015	11,563.948	11,563.948	11,563.948	0.000	0.000	0.000
01/06/2015	30/06/2015	4,735.701	4,735.701	4,735.701	0.670	0.670	0.677
01/07/2015	31/07/2015	1,926.892	1,926.892	1,926.892	0.000	0.000	0.000
01/08/2015	31/08/2015	3,080.514	3,080.514	3,080.514	0.000	0.000	0.000
01/09/2015	30/09/2015	4,784.169	4,784.169	4,784.169	0.000	0.000	0.000
01/10/2015	31/10/2015	11,033.792	11,033.792	11,033.792	0.571	0.571	0.577
01/11/2015	30/11/2015	6,840.015	6,840.015	6,840.015	0.000	0.000	0.000
01/12/2015	31/12/2015	10,148.018	10,148.018	10,148.018	0.000	0.000	0.000
2015 Subtotal		-	-	95,853.045	-	-	1.253
01/01/2016	31/01/2016	6,079.787	6,079.787	6,079.787	0.000	0.000	0.000
01/02/2016	29/02/2016	10,333.084	10,333.084	10,333.084	0.230	0.230	0.232
01/03/2016	31/03/2016	11,201.073	11,201.073	11,201.073	0.000	0.000	0.000
01/04/2016	30/04/2016	10,226.520	10,226.520	10,226.520	0.000	0.000	0.000
01/05/2016	31/05/2016	13,308.354	13,308.354	13,308.354	0.000	0.000	0.000
01/06/2016	30/06/2016	4,397.406	4,397.406	4,397.406	0.000	0.000	0.000
01/07/2016	31/07/2016	2,629.447	2,629.447	2,629.447	0.000	0.000	0.000
01/08/2016	31/08/2016	2,878.811	2,878.811	2,878.811	0.000	0.000	0.000
01/09/2016	30/09/2016	3,997.546	3,997.546	3,997.546	0.000	0.000	0.000
01/10/2016	31/10/2016	10,086.042	10,086.042	10,086.042	0.980	0.980	0.990
01/11/2016	30/11/2016	15,786.432	15,786.432	15,786.432	0.000	0.000	0.000
01/12/2016	31/12/2016	12,904.051	12,904.051	12,904.051	0.000	0.000	0.000
2016 Subtotal		-	-	103,828.553	-	-	1.222
Total (01/05/2012-31/12/2016)		-	-	484,828.431	-	-	9.414

Table 4 Monitored values of electricity imported from the grid through the main line (EG_{im})

EG _{im} (MWh)				
Period		Monthly readings from M1	Values from sales receipts**	Conservative Quantity
From	To	D	E	F= Max (D, E)

04/04/2012	03/05/2012*	19.503	43.773	43.773
04/05/2012	04/06/2012	24.255	54.439	54.439
05/06/2012	04/07/2012	31.284	70.215	70.215
05/07/2012	04/08/2012	54.648	122.654	122.654
05/08/2012	03/09/2012	45.936	103.101	103.101
04/09/2012	04/10/2012	54.945	123.321	123.321
05/10/2012	04/11/2012	11.781	26.442	26.442
05/11/2012	03/12/2012	3.366	7.555	7.555
04/12/2012	31/12/2012	11.979	26.886	26.886
2012 Subtotal		-	-	534.613
01/01/2013	03/02/2013	14.454	32.441	32.441
04/02/2013	04/03/2013	10.098	22.664	22.664
05/03/2013	02/04/2013	11.583	25.997	25.997
03/04/2013	05/05/2013	45.540	102.212	102.212
06/05/2013	03/06/2013	20.493	45.995	45.995
04/06/2013	03/07/2013	53.262	119.544	119.544
04/07/2013	03/08/2013	49.599	111.322	111.322
04/08/2013	24/09/2013	16.830	37.774	37.774
25/09/2013	04/10/2013	7.920	17.776	17.776
05/10/2013	07/11/2013	22.671	50.884	50.884
08/11/2013	09/12/2013	6.633	14.887	14.887
10/12/2013	31/12/2013	17.820	39.996	39.996
2013 Subtotal		-	-	621.492
01/01/2014	08/02/2014	16.929	37.996	37.996
09/02/2014	09/03/2014	36.531	81.992	81.992
10/03/2014	08/04/2014	37.521	84.214	84.214
09/04/2014	08/05/2014	31.779	71.326	71.326
09/05/2014	08/06/2014	22.770	51.106	51.106
09/06/2014	09/07/2014	53.460	119.988	119.988
10/07/2014	07/08/2014	38.610	86.658	86.658
08/08/2014	08/09/2014	49.500	111.100	111.100
09/09/2014	08/10/2014	49.500	111.100	111.100

09/10/2014	08/11/2014	25.740	57.772	57.772
09/11/2014	08/12/2014	5.940	13.332	13.332
09/12/2014	31/12/2014	7.920	17.776	17.776
2014 Subtotal		-	-	844.360
01/01/2015	08/02/2015	25.740	57.772	57.772
09/02/2015	08/03/2015	14.850	33.330	33.330
09/03/2015	08/04/2015	19.800	44.440	44.440
09/04/2015	08/05/2015	25.740	57.772	57.772
09/05/2015	08/06/2015	13.860	31.108	31.108
09/06/2015	08/07/2015	53.460	119.988	119.988
09/07/2015	08/08/2015	54.450	122.210	122.210
09/08/2015	08/09/2015	87.120	195.536	195.536
09/09/2015	08/10/2015	32.670	73.326	73.326
09/10/2015	08/11/2015	34.650	77.770	77.770
09/11/2015	08/12/2015	36.630	82.214	82.214
09/12/2015	31/12/2015	14.751	33.108	33.108
2015 Subtotal		-	-	928.574
01/01/2016	08/01/2016	0.000	11.332	11.332
09/01/2016	31/01/2016	19.899	33.330	33.330
01/02/2016	06/03/2016	27.819	62.438	62.438
07/03/2016	06/04/2016	32.175	72.215	72.215
07/04/2016	06/05/2016	43.956	98.657	98.657
07/05/2016	06/06/2016	20.889	46.884	46.884
07/06/2016	06/07/2016	51.381	115.322	115.322
07/07/2016	06/08/2016	32.076	71.993	71.993
07/08/2016	06/09/2016	24.354	54.661	54.661
07/09/2016	06/10/2016	50.490	113.322	113.322
07/10/2016	06/11/2016	30.294	67.993	67.993
07/11/2016	06/12/2016	17.523	39.329	39.329
07/12/2016	31/12/2016	4.059	9.110	9.110
2016 Subtotal		-	-	787.476
Total (01/05/2012-31/12/2016)		-	-	3,716.515

Note:

*The electricity imported from the main line for the three days from 01-May-2012 to 03-May-2012 was settled in the reading records and sales receipts of the period from 04-April-2012 to 03-May-2012. To be conservative, the electricity imported for the whole period from 04-April-2012 to 03-May-2012 was included in the emission reductions calculation.

**The readings of EG_{im} are sourced from the monthly reading records and due to that the EG_{im} for the project was settled together with another VCS project (VCS ID 395) in the same monthly sales receipts, to be conservative, all the imported electricity for the two projects on sales receipts was included for ER calculations.

Table 5 Calculations of EG_y

Period	EG_{ex} (MWh)	EG_{im} (MWh)	EG_{backup} (MWh)	EG_y (MWh)
-	C	F	I	J=C-F-I
01/05/2012-31/12/2012	65,965.896	534.613	1.576	65,429.707
01/01/2013-31/12/2013	118,743.878	621.492	3.232	118,119.154
01/01/2014-31/12/2014	100,437.059	844.360	2.131	99,590.568
01/01/2015-31/12/2015	95,853.045	928.574	1.253	94,923.218
01/01/2016-31/12/2016	103,828.553	787.476	1.222	103,039.855
Total	484,828.431	3,716.515	9.414	481,102.502

5.2 Project Emissions

As per the applied methodology, the project emission is zero.

5.3 Leakage

As per the applied methodology, leakage of the project is not considered.

5.4 Net GHG Emission Reductions and Removals

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2012 (01-May-2012 to 31-Dec-2012)	74,975	0	0	74,975
2013	135,352	0	0	135,352
2014	114,120	0	0	114,120
2015	108,772	0	0	108,772

2016	118,073	0	0	118,073
Total	551,292	0	0	551,292

Comparison of actual emission reduction and the estimation in the registered PD.

Monitoring Period	Emission reduction achieved (tCO ₂)	Estimated ER in approved revised CDM PDD (tCO ₂)	Difference
01-May-2012 to 31-Dec-2016	551,292	667,668 ³	-17.43%

The actual emission reduction achieved during this monitoring period is 17.43%⁴ lower than the estimation for the same period in the approved revised CDM PDD mainly due to the fluctuation in wind resources. Therefore, the emission reductions during this monitoring period are deemed to be reasonable.

³ As per the registered PDD, the annual emission reduction of the project is 142,848 tCO₂, and it is covering 1,706 days from 01/05/2012 to 31/12/2016, thus the expected emission reduction in the monitoring period is 667,668 tCO₂ (calculated as 142,848/365d*1,706d).

⁴ (551,292 - 667,668) tCO₂ / 667,668 tCO₂ * 100% = 17.43%